

ECSE 4560/ 6560

Modern Communication Systems

Lecture 18: MIMO Part3: STBC Diversity

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **STBC Diversity**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Friday 11/7**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. HW6 (MIMO code) due 11/14
 - Exam 1: Friday Oct 24 Graded (Submit regrading request by Tue **11/4**)
 - Exam 2: **Friday Nov 21**

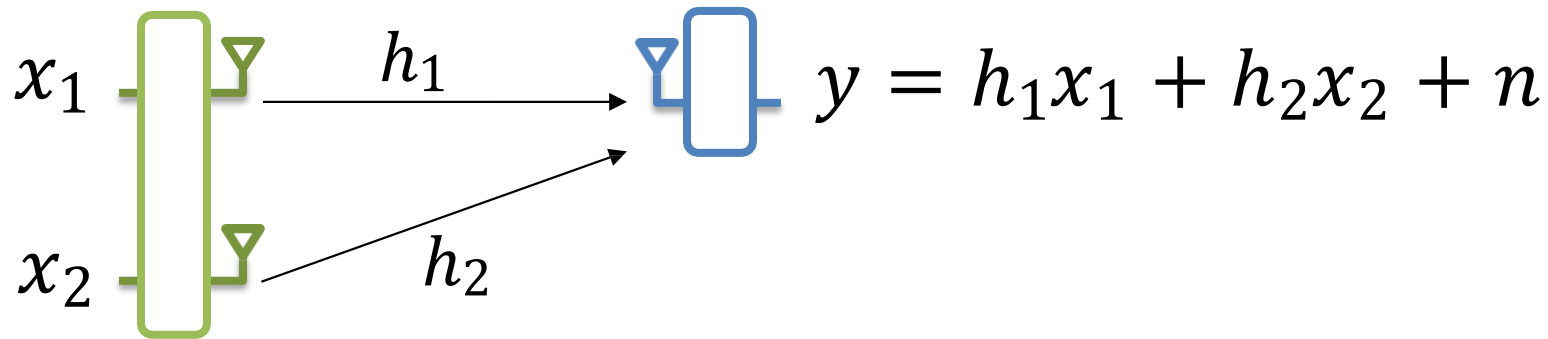
Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**

Today's Class

- MIMO Gains
 - Receiver Diversity
 - **Transmitter Diversity**
 - **Space-Time codes (Alamouti Codes)**
- MIMO Multiplexing Gains
 - Zero-forcing
 - SVD
 - MU MIMO

Transmitter Diversity



What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

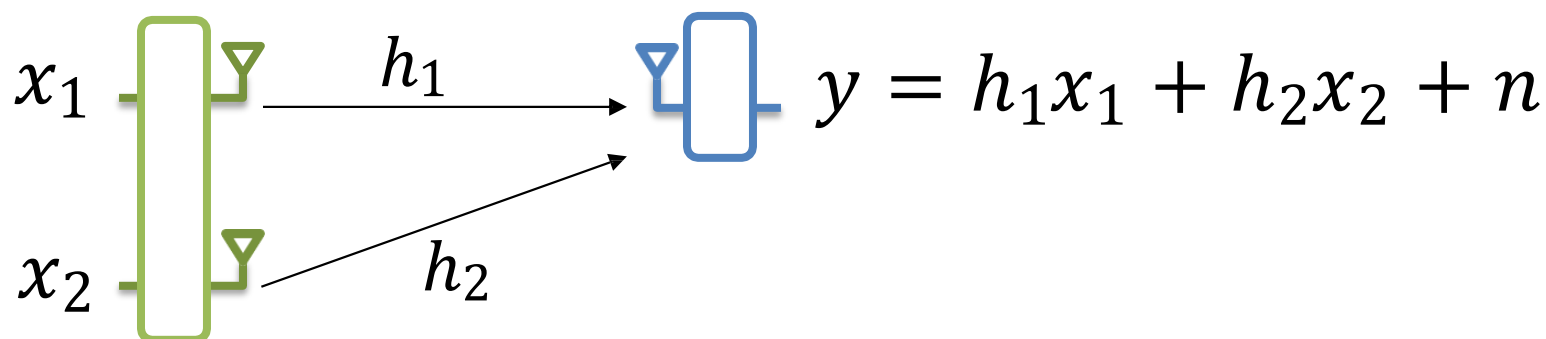
$$y = (h_1 + h_2)x + n$$

Channels can sum up destructively! $h_1 + h_2 \approx 0$

Total transmit power = $2P \rightarrow$ Doubled TX power

$\rightarrow$ Why not use 1 TX with $2P$

Transmitter Diversity

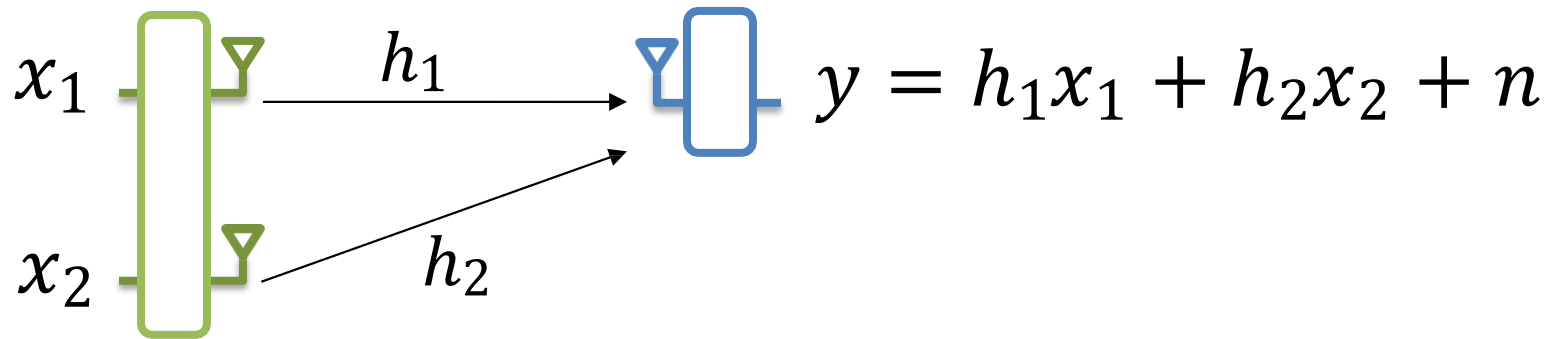


What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

- Must ensure signals sum up constructively.
- Must ensure total TX power = $E[|x_1|^2] + E[|x_2|^2]$
 $= E[|x|^2] = P$

Transmitter Diversity



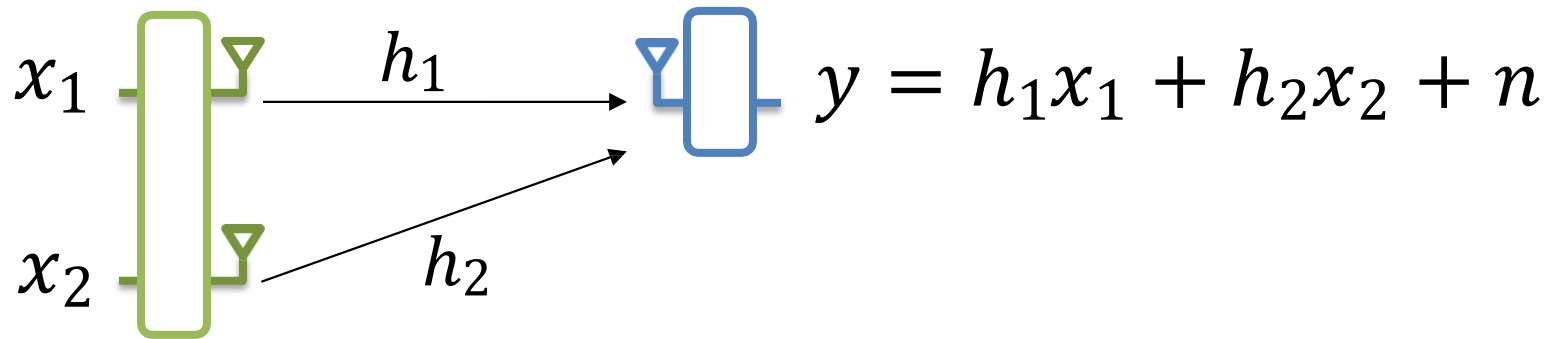
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$\left. \begin{array}{l} x_1 = \alpha_1 x \\ x_2 = \alpha_2 x \end{array} \right\} y = (\alpha_1 h_1 + \alpha_2 h_2)x + n$$

$$\begin{aligned} \text{Set: } \alpha_1 = h_1^*, \alpha_2 = h_2^* &\rightarrow y = (h_1^* h_1 + h_2^* h_2)x + n \\ &= (|h_1|^2 + |h_2|^2)x + n \end{aligned}$$

Transmitter Diversity



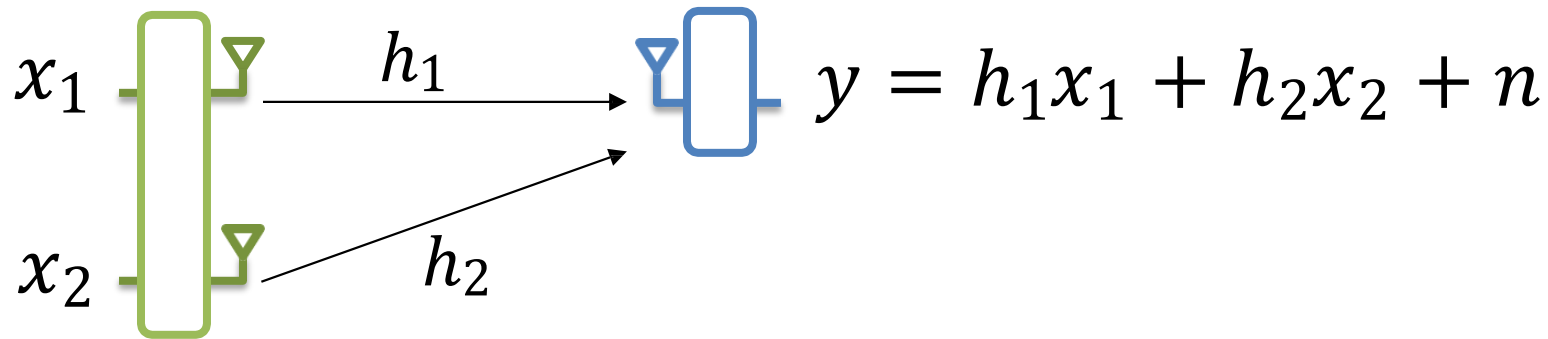
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$\begin{aligned} \text{Total TX Power} &= \mathbb{E}[|x_1|^2] + \mathbb{E}[|x_2|^2] \\ &= \frac{|h_1|^2}{|h_1|^2 + |h_2|^2} P + \frac{|h_2|^2}{|h_1|^2 + |h_2|^2} P = P \end{aligned}$$

Transmitter Diversity



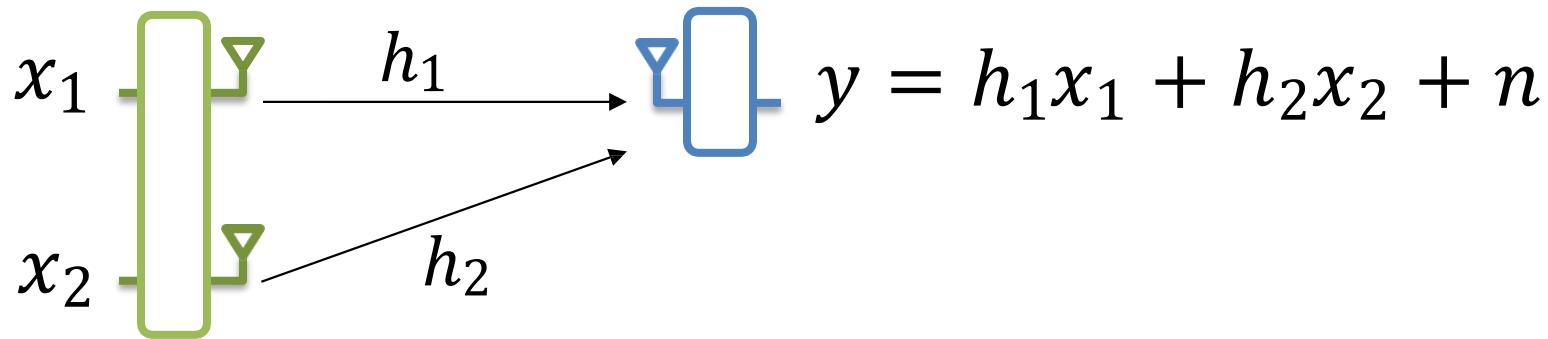
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{h_1 h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + \frac{h_2 h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + n$$

Transmitter Diversity



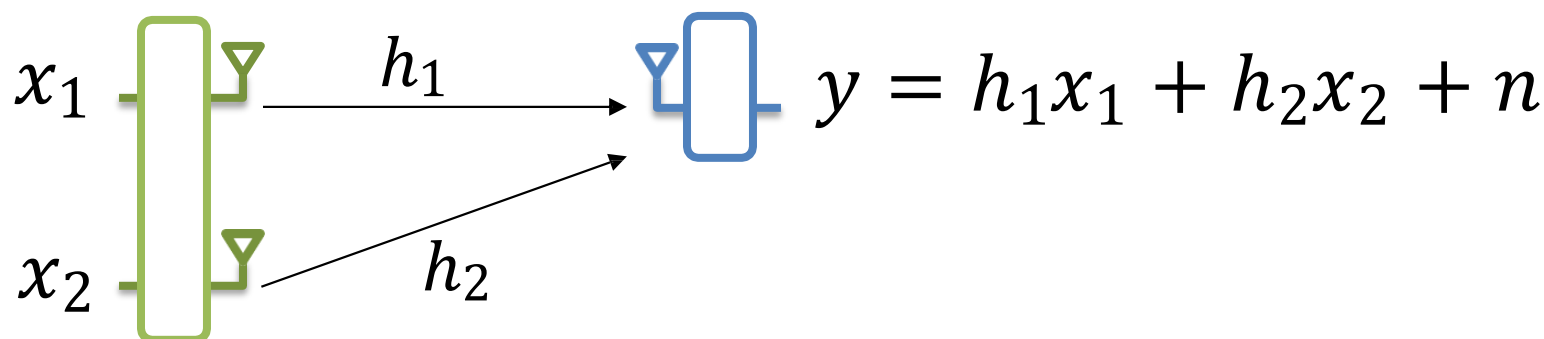
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{|h_1|^2 + |h_2|^2}{\sqrt{|h_1|^2 + |h_2|^2}} x + n = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

Transmitter Diversity



What should we transmit on each antenna?

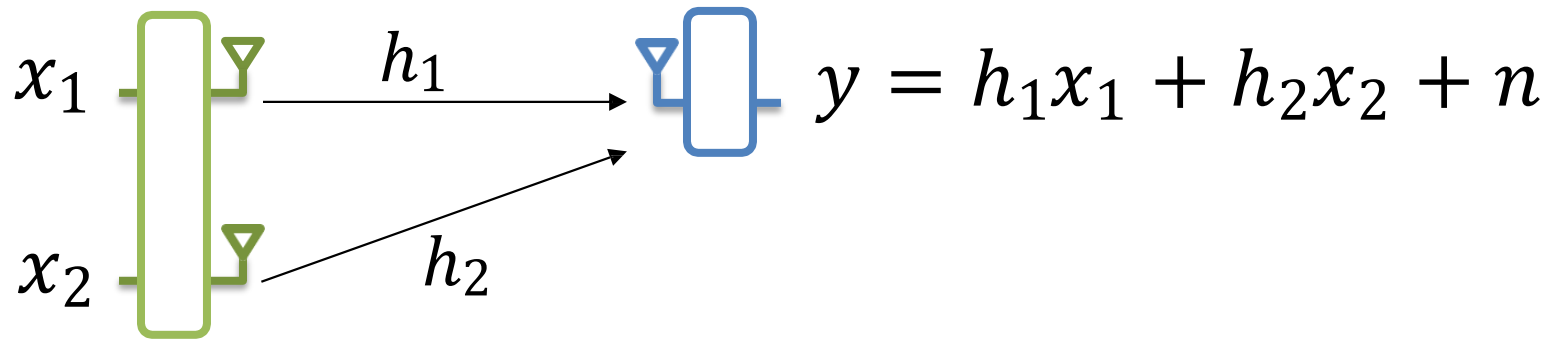
Option 2: Maximum Ratio Combining (MRC)

$$y = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

$$SNR = (|h_1|^2 + |h_2|^2) \frac{P}{\sigma^2} \rightarrow \text{Similar SNR Gain to RX Diversity}$$

Caveat: MRC at TX Requires Feedback from the Receiver!

Transmitter Diversity – HW5



Consider narrowband flat fading with channel vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \in \mathbb{C}^2, \quad n \sim \mathcal{CN}(0, N_0), \quad \mathbb{E}[|x|^2] = P.$$

Unless stated otherwise, total TX power per channel use is P (shared across the two antennas).

(a) **MRC beamforming with CSIT (derivation).** With perfect Channel state information at the transmitter (CSIT), the transmitter applies a unit-norm precoder $\mathbf{w} \in \mathbb{C}^2$ and sends $\mathbf{x} = \mathbf{w}x$.

(i) Under $\|\mathbf{w}\|_2 = 1$, derive $\mathbf{w}$ that maximizes instantaneous receive SNR.

(ii) Show that the resulting SNR equals $\gamma_{\text{BF}} = \frac{\|\mathbf{h}\|_2^2 P}{N_0} = \frac{(|h_1|^2 + |h_2|^2)P}{N_0}$.

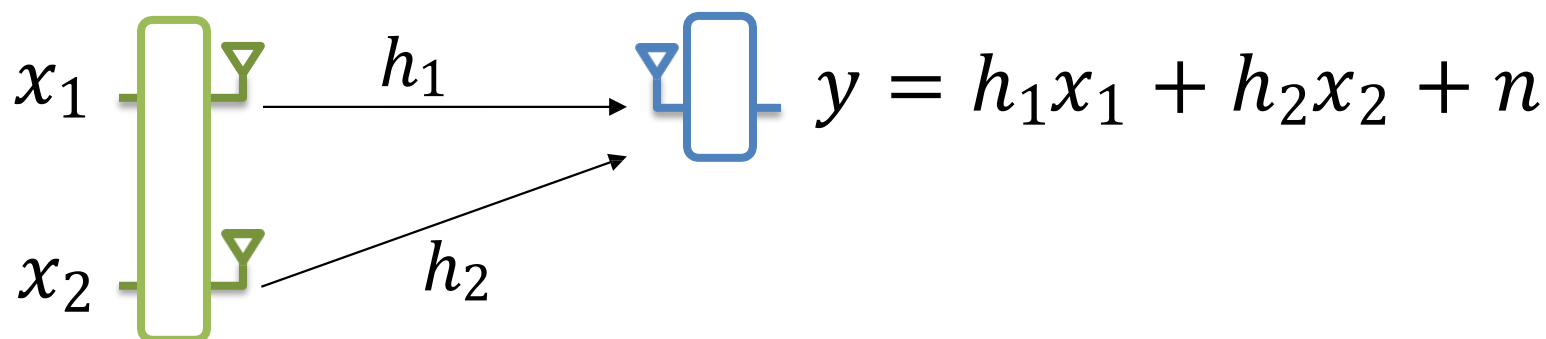
Transmitter Diversity – HW5

Transmitter Diversity – HW5

Today's Class

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 - SVD
 - MU MIMO

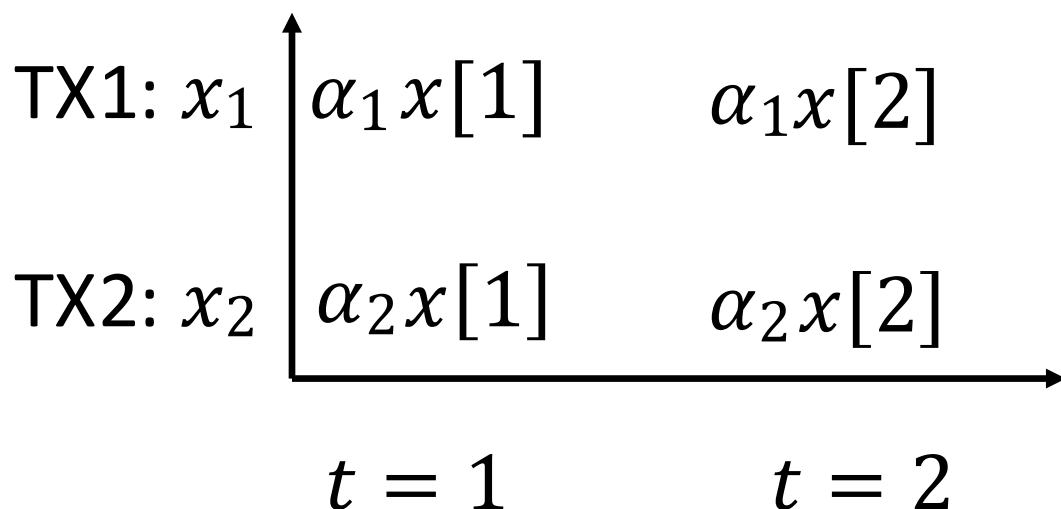
Transmitter Diversity



What should we transmit on each antenna?

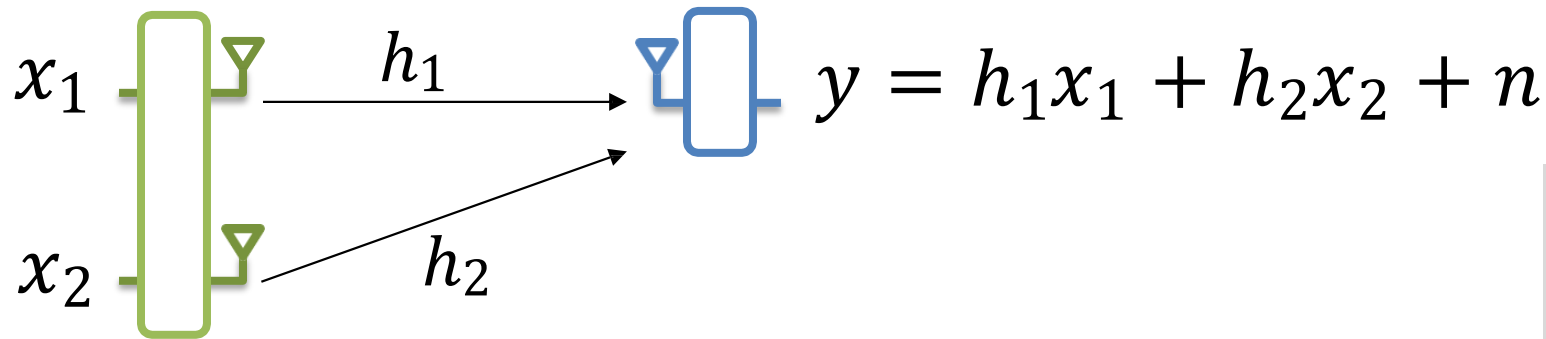
Solution: Use Space-Time Codes

MRC codes across space only (Requires Channel Feedback):



$$\alpha_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$
$$\alpha_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$

Transmitter Diversity



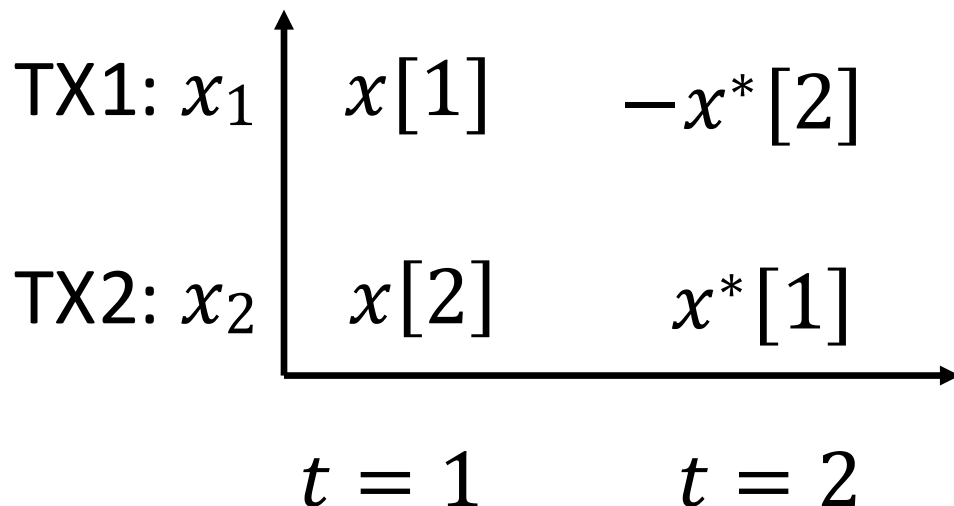
What should we transmit on each antenna?

Solution: Use Space-Time Codes



Siavash Alamouti
1996 AT&T Labs

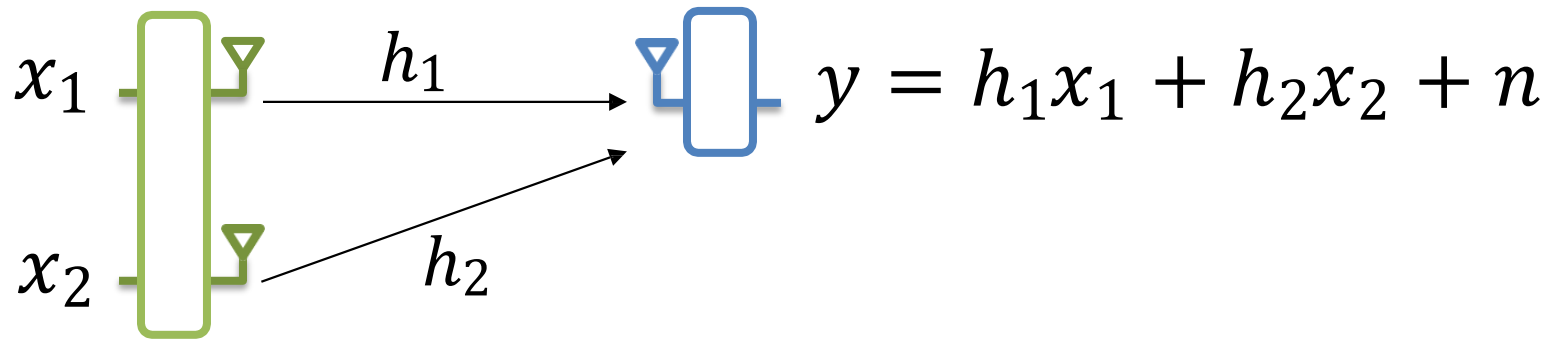
Alamouti Codes:



$$y[1] = h_1x[1] + h_2x[2]$$

$$y[2] = -h_1x^*[2] + h_2x^*[1]$$

Transmitter Diversity



What should we transmit on each antenna?

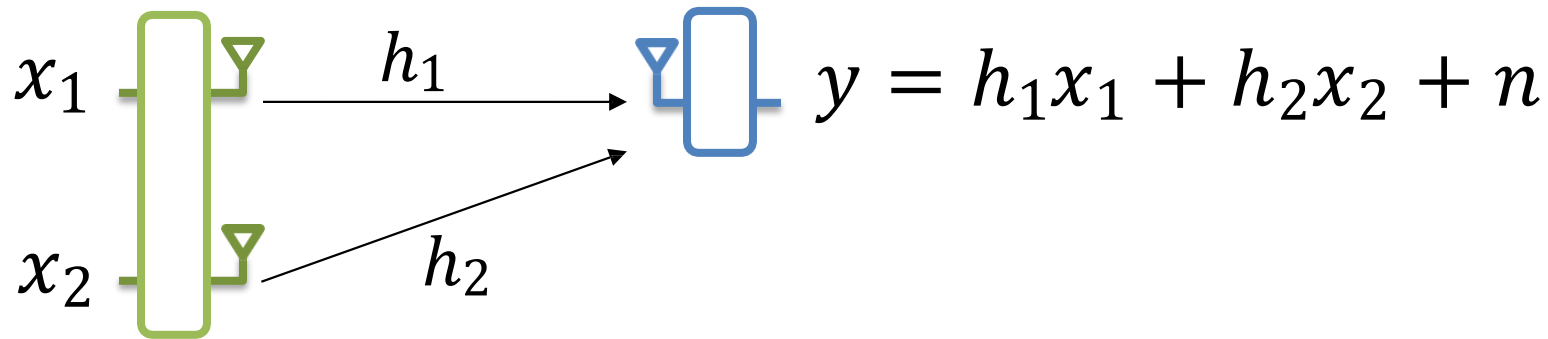
Solution: Use Space-Time Codes

Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

$$\begin{aligned} h_1^*y[1] + h_2y^*[2] &= h_1^*h_1x[1] + h_1^*h_2x[2] \\ &\quad - h_2h_1^*x[2] + h_2h_2^*x[1] \end{aligned}$$

Transmitter Diversity



What should we transmit on each antenna?

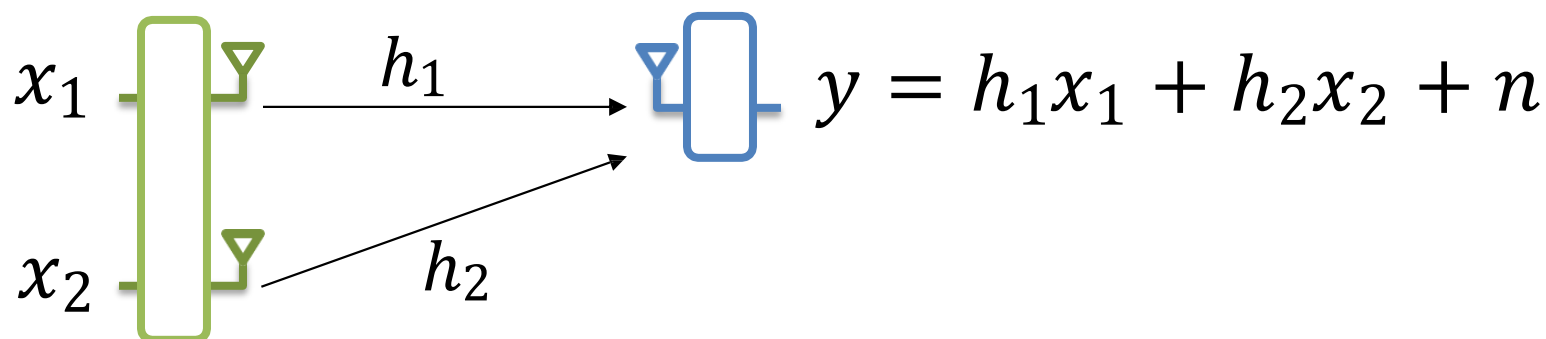
Solution: Use Space-Time Codes

Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

$$\begin{aligned} h_2^*y[1] - h_1y^*[2] &= \cancel{h_2^*h_1x[1]} + h_2^*h_2x[2] \\ &\quad + h_1h_1^*x[2] - \cancel{h_1h_2^*x[1]} \\ &= (|h_1|^2 + |h_2|^2)x[2] \end{aligned}$$

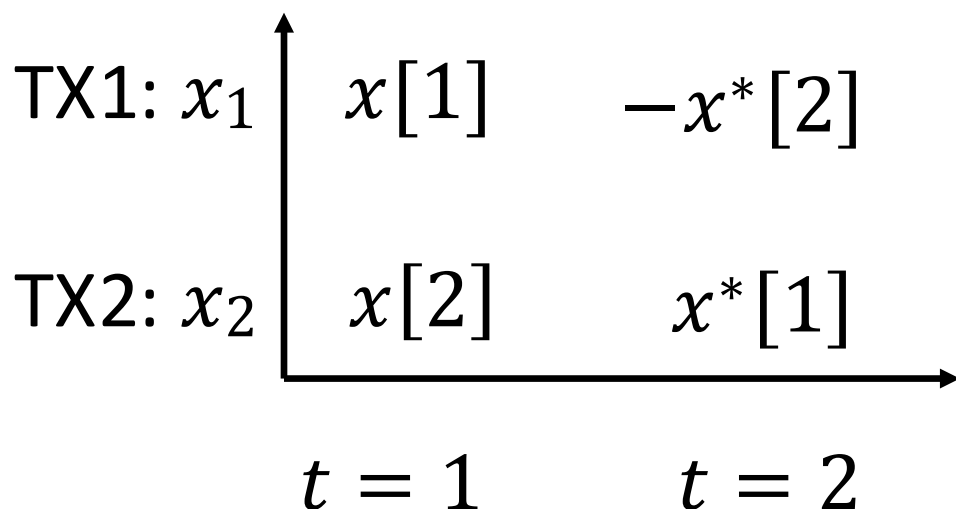
Transmitter Diversity



What should we transmit on each antenna?

Solution: Use Space-Time Codes

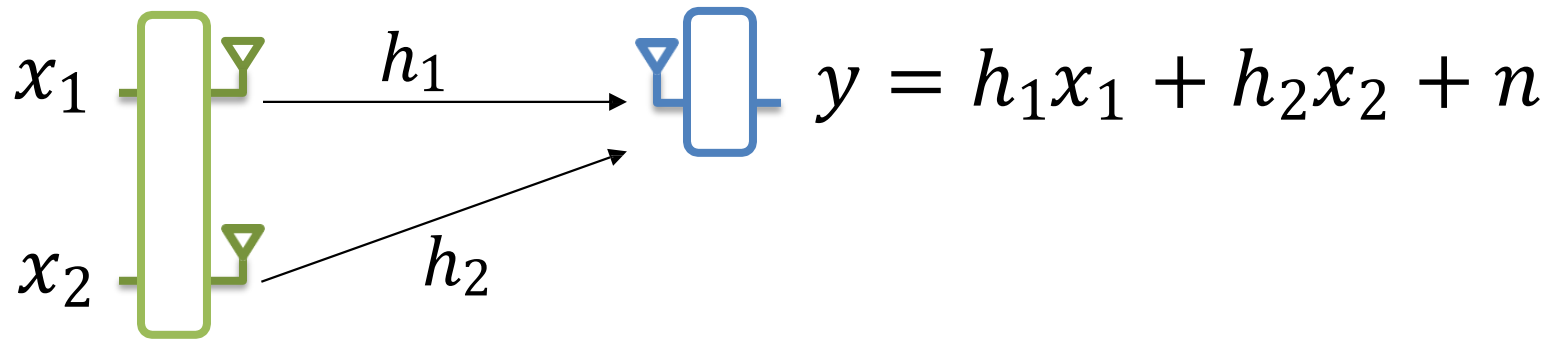
Alamouti Codes:



$$h_1^*y[1] + h_2y^*[2] = (|h_1|^2 + |h_2|^2)x[1]$$

$$h_2^*y[1] - h_1y^*[2] = (|h_1|^2 + |h_2|^2)x[2]$$

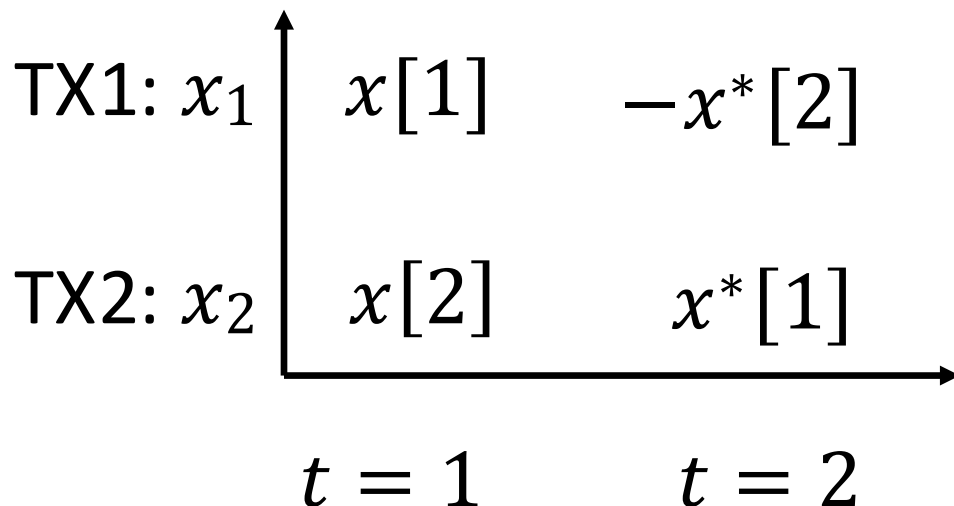
Transmitter Diversity - Normalize



What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:



$$h_1^* y[1] + h_2 y^*[2] = (|h_1|^2 + |h_2|^2) x[1]$$

$$h_2^* y[1] - h_1 y^*[2] = (|h_1|^2 + |h_2|^2) x[2]$$

Alamouti SNR

(c) **Alamouti STBC (no CSIT)**. Using the 2×1 Alamouti code over two time slots:

$$\text{slot } t : [x_1 \ x_2], \quad \text{slot } t + 1 : [-x_2^* \ x_1^*].$$

- (i) Write the two received samples y_t and y_{t+1} in terms of $h_1, h_2, x_1, x_2, n_t, n_{t+1}$.
- (ii) Form the standard two-equation linear combiner that decouples (x_1, x_2) and show that each symbol experiences an effective gain $|h_1|^2 + |h_2|^2$ (assume total TX power per *time slot* equals P and the usual Alamouti normalization).
- (iii) State the resulting per-symbol SNR γ_{Alam} in terms of $P, N_0, |h_1|^2, |h_2|^2$ and the code rate.

Alamouti SNR

$$\begin{aligned}\text{SNR} &= \frac{\|\mathbf{h}\|^2}{\sigma_n^2} \text{E} \left\{ |x_1|^2 \right\} \\ &= \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1,\end{aligned}$$

$$\frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1, \quad \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_2,$$

$$P_1 = P_2 = \frac{P}{2}.$$

$$\text{SNR} = \frac{P}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} = \frac{1}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P$$

Alamouti SNR across two timeslots

Code Rate

If an STBC transmits k independent information symbols over T time slots, the **code rate** R is defined as:

$$R = \frac{k}{T} \text{ (symbols per channel use)}$$

Only the Alamouti code (for 2 transmit antennas) achieves a full code rate of 1 while maintaining full orthogonality and full diversity in complex constellations.

MIMO Gains

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(SNR/N)$

Diversity Gain:

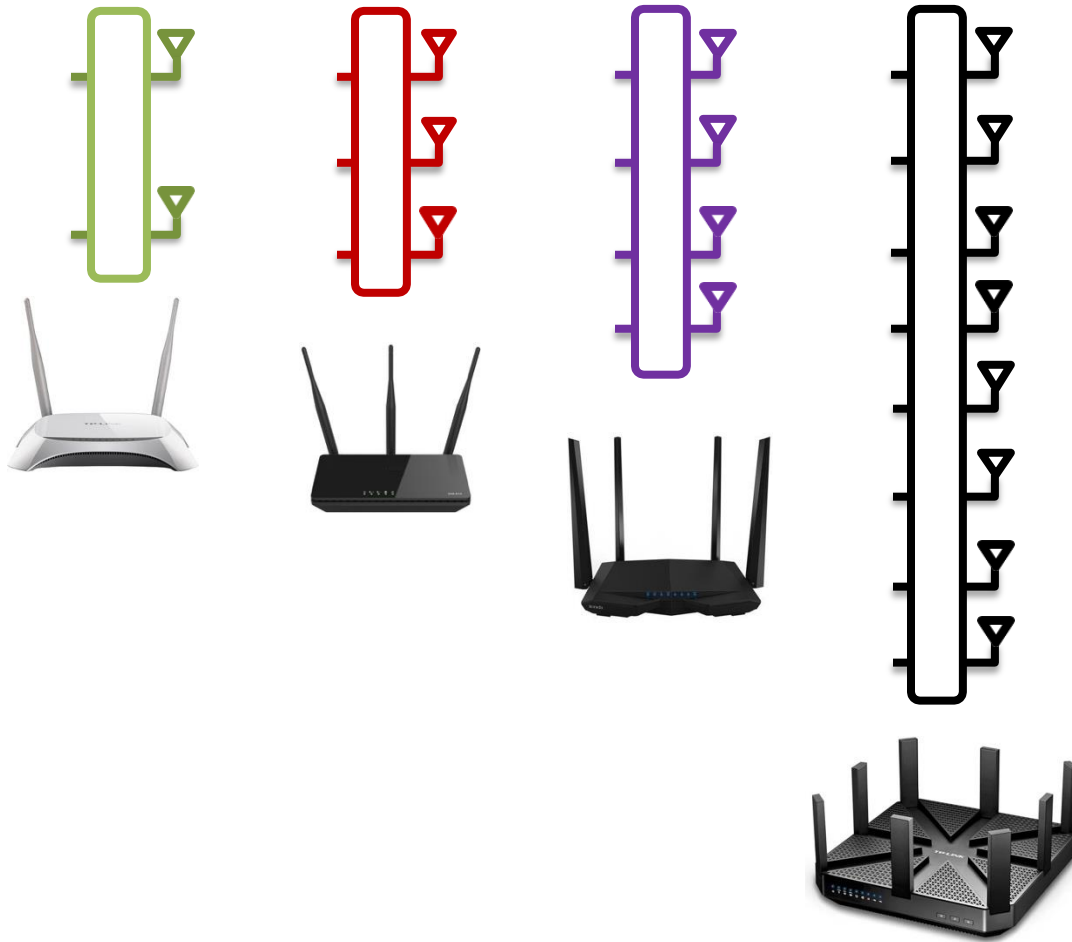
- Increase SNR of the received packets
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(SNR \times N^2)$

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MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO

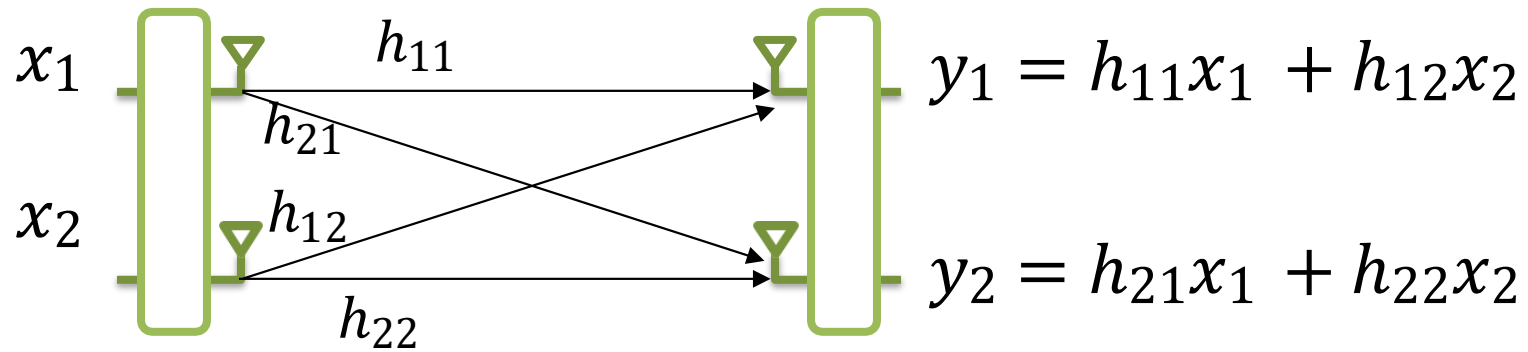


WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

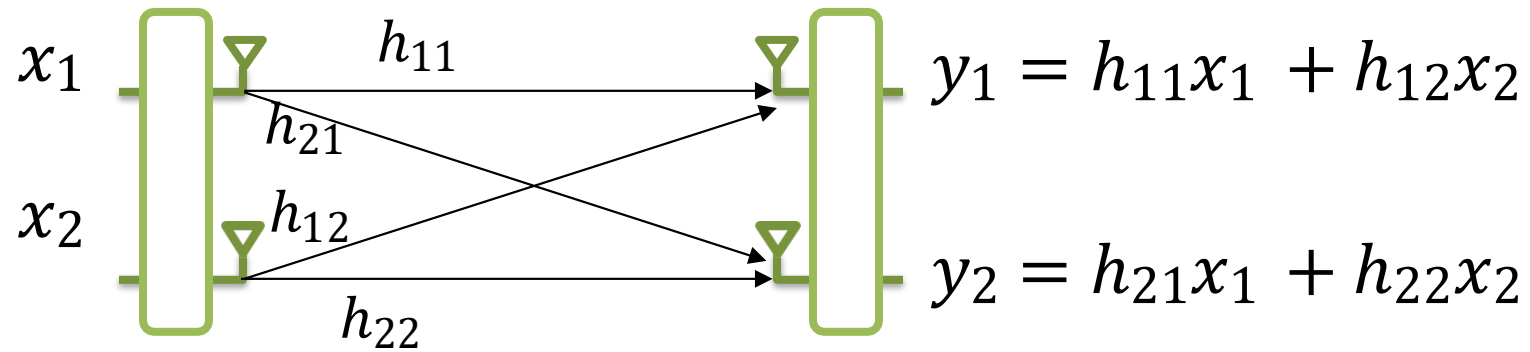
MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

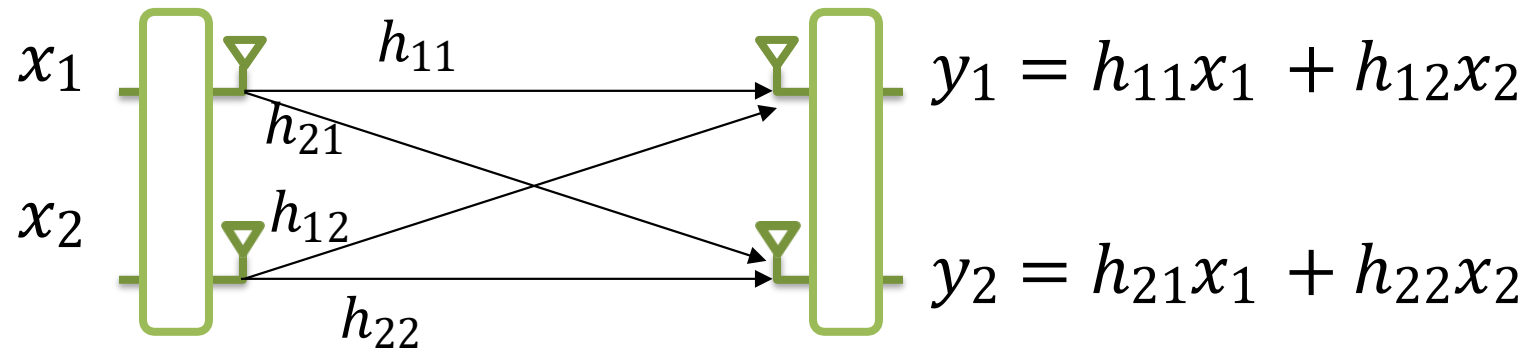
$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Zero-forcing based MIMO



$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

SVD-based MIMO



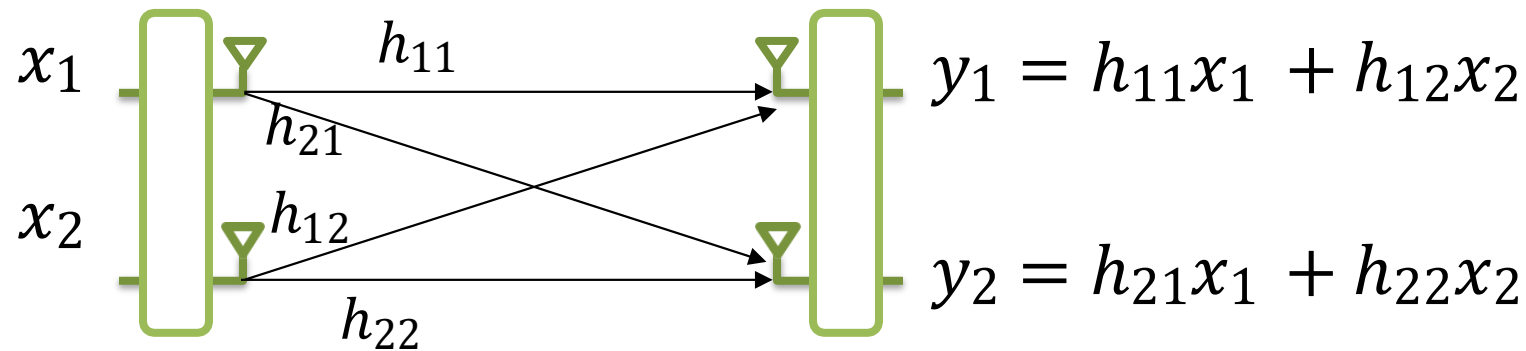
$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1, 1 \leq i \leq t$$

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0, i \neq j, 1 \leq i, j \leq t$$

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

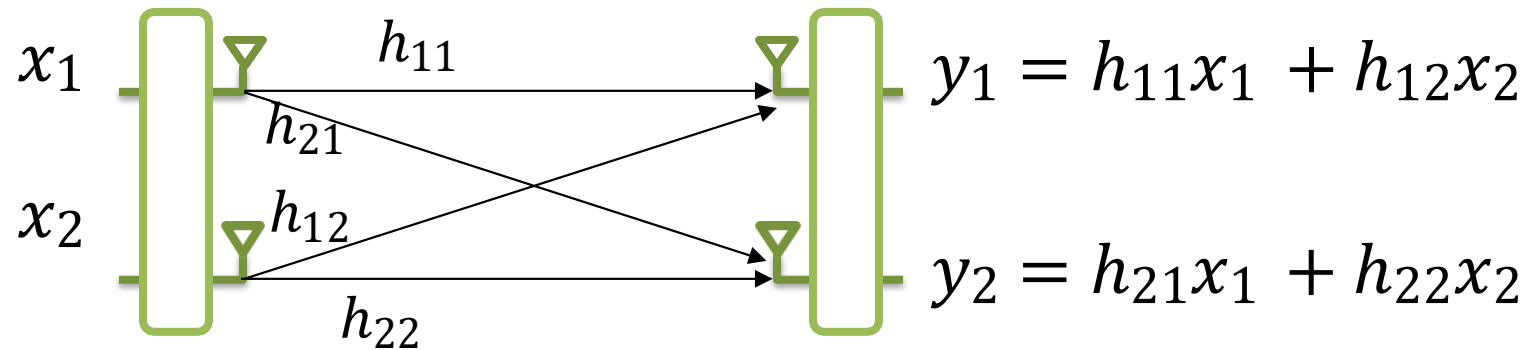
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} \vdots \\ \vdots \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{y}} = \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

$$\mathbf{y} = \underbrace{\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H}_{\mathbf{H}}\mathbf{x} + \mathbf{n}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{V}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

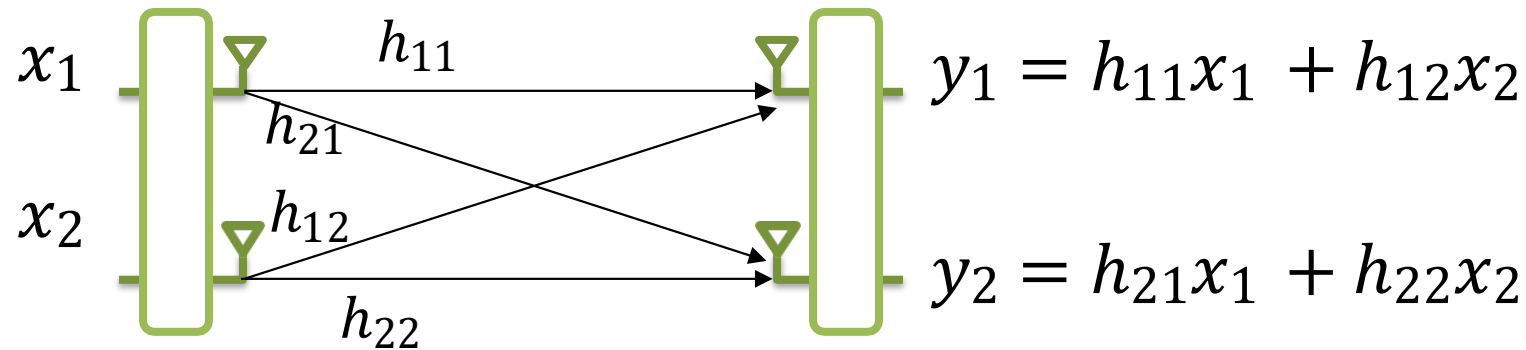
$$\underbrace{\mathbf{U}^H\mathbf{y}}_{\tilde{\mathbf{y}}} = \mathbf{U}^H (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \mathbf{n})$$

$$= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{y}} = \underbrace{\mathbf{U}^H\mathbf{U}}_{\mathbf{I}_t}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \underbrace{\mathbf{U}^H\mathbf{n}}_{\tilde{\mathbf{n}}}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

SVD-based MIMO



$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

$$\vdots$$

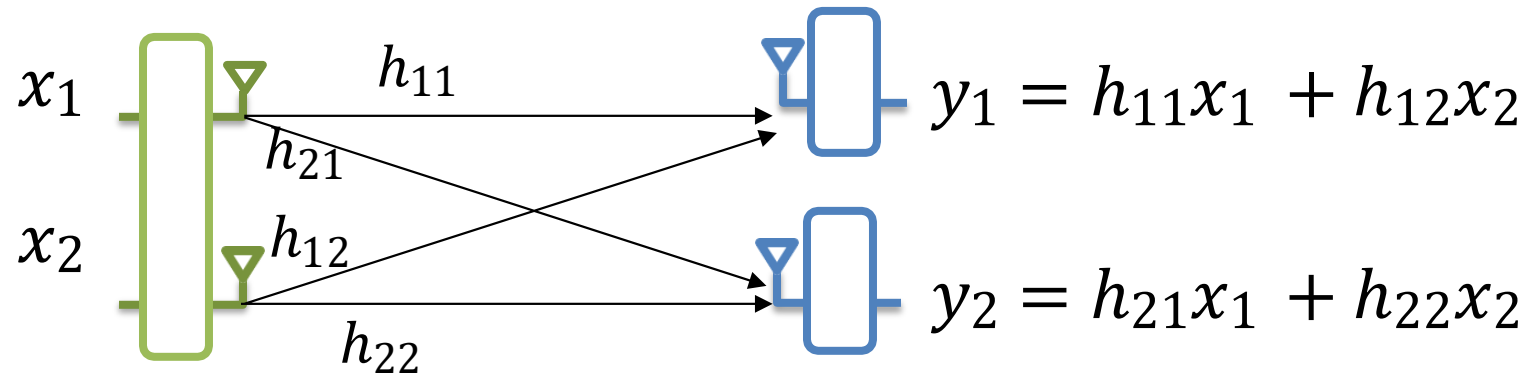
$$\tilde{y}_t = \sigma_t \tilde{x}_t + \tilde{n}_t$$

$$C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$

Multi-User MIMO

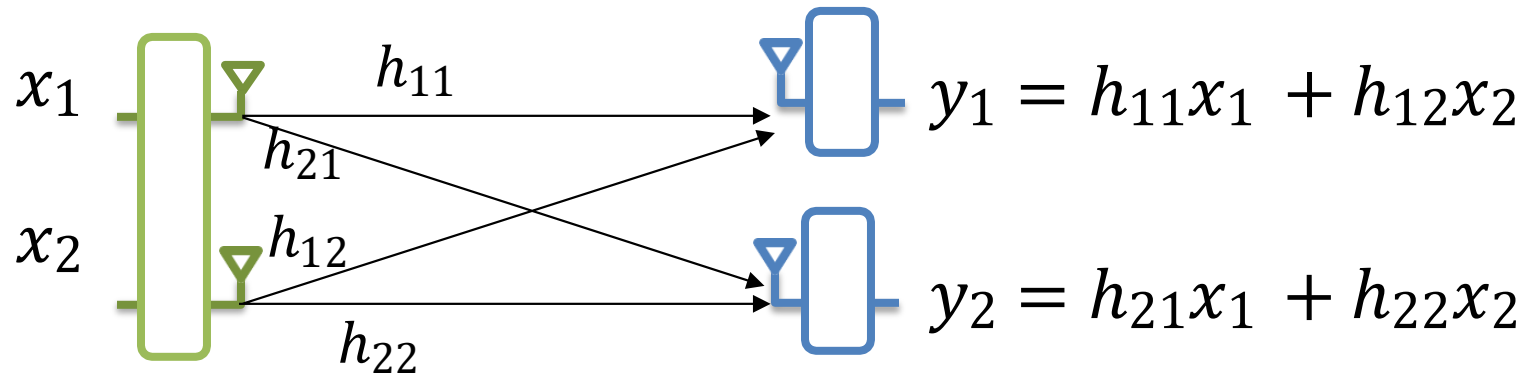


$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Does not work. Receivers do not have access to each other's signals

Precoding



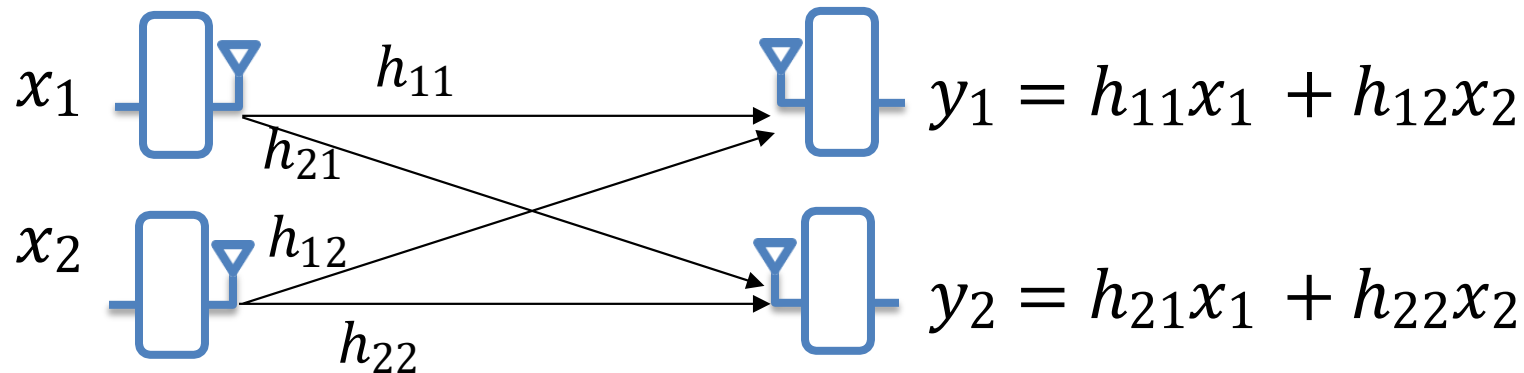
Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Also known as Beamforming or Zero-Forcing

Requires Channel Feedback!

What about distributed transmitters?



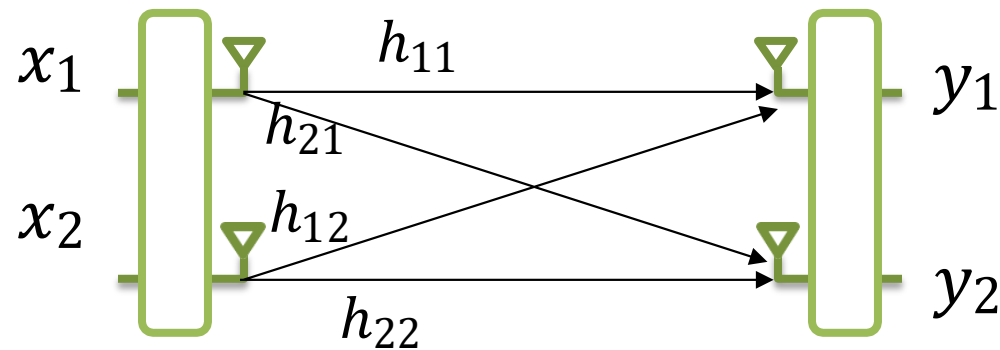
Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Does not work. Transmitters do not have access to each other's signals

False: what if transmitters are APs connected over Ethernet?

Clock Synchronization in MIMO

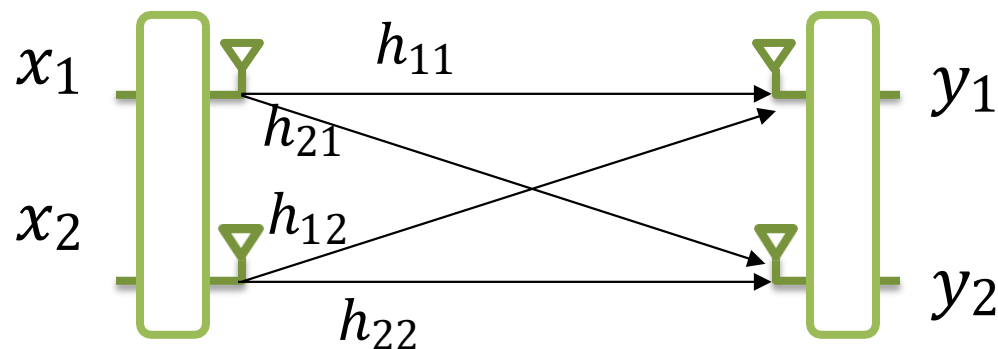


$$y_1 = h_{11}x_1e^{-j2\pi\Delta f_{11}t} + h_{12}x_2e^{-j2\pi\Delta f_{12}t}$$

$$y_2 = h_{21}x_1e^{-j2\pi\Delta f_{21}t} + h_{22}x_2e^{-j2\pi\Delta f_{22}t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

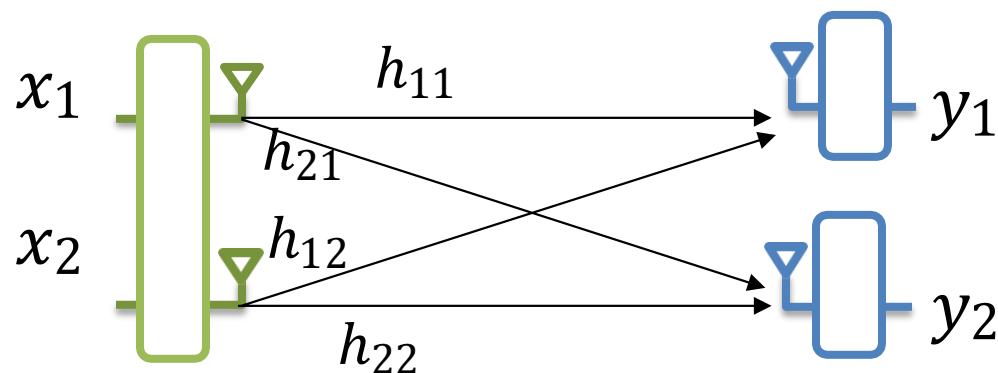


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

On chip MIMO: $\Delta f_{11} = \Delta f_{21} = \Delta f_{12} = \Delta f_{22} = \Delta f$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} e^{-j2\pi\Delta f t} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

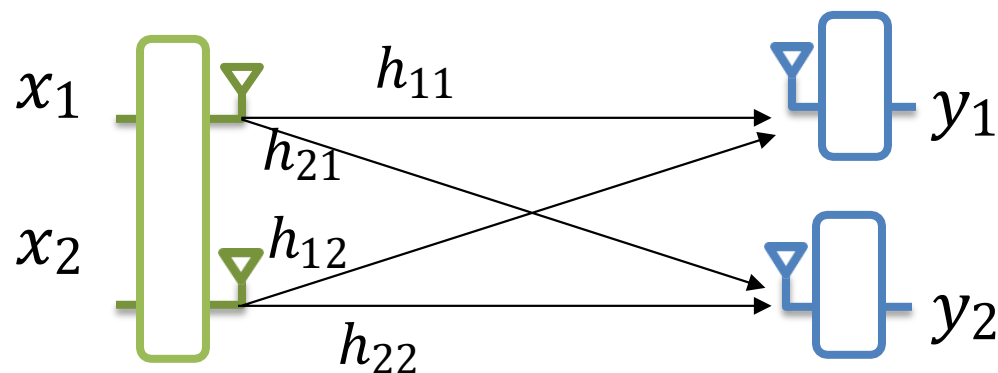


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_1t} & h_{12} e^{-j2\pi\Delta f_1t} \\ h_{21} e^{-j2\pi\Delta f_2t} & h_{22} e^{-j2\pi\Delta f_2t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

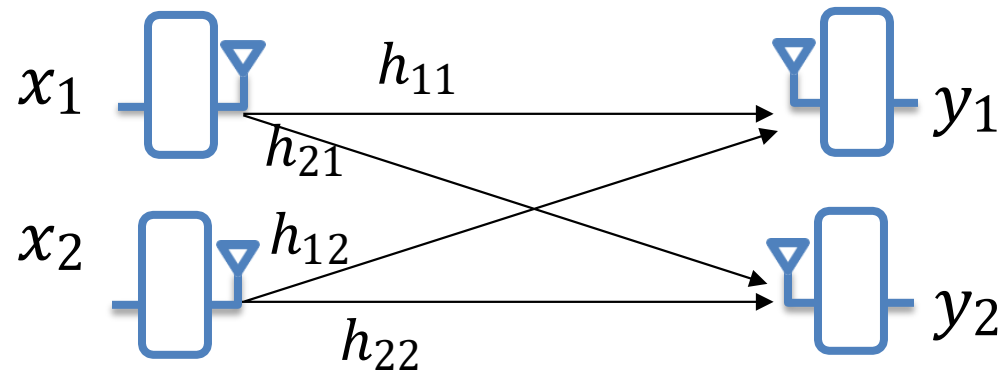


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e^{-j2\pi\Delta f_1 t} & 0 \\ 0 & e^{-j2\pi\Delta f_2 t} \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Distributed MIMO: $\Delta f_{11} \neq \Delta f_{12} \neq \Delta f_{21} \neq \Delta f_{22}$